THARUNKUMAR K

FULL STACK DEVELOPER | AI SYSTEMS ENGINEER | SOFTWARE DEVELOPMENT ENGINEER

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PROFESSIONAL SUMMARY

Full Stack Developer and Systems Builder experienced in engineering scalable web applications, distributed systems, and AI-powered platforms using React 19, Node.js, TypeScript, Java 17, and Python. Interned at Neura Global and Infosys Springboard delivering production software. Creator of VSBEC IT Vault (deployed to 365+ daily students), GOAT Collaborative IDE (1st Place National Winner), and Smart Helmet IoT (SIH Top 50 Nominee). Active problem solver with deep CS fundamentals.

TECHNICAL SKILLS & CS FUNDAMENTALS

Languages & Core CS:	C, C++, Java 17, Python, TypeScript, JavaScript, SQL, Data Structures & Algorithms
Systems & Architecture:	P2P WebRTC Mesh, Socket.IO, Docker Sandbox Jails, REST APIs, Microservices, Spring Boot
AI Driving & LLMs:	Google Gemini API, Ollama (Local LLaMA 3), ChromaDB (Vector RAG), Prompt Architecture
Databases & Cloud:	PostgreSQL, Supabase (RLS), MongoDB, SQLite, Cloudinary, Docker, Git, GitHub Actions, Vercel
Frontend & UI Systems:	React 19, Next.js 15, Vite, Tailwind CSS, Monaco Editor Core, State Machines, WebSockets

EDUCATION

Bachelor of Technology (B.Tech) in IT (CGPA: 8.5 / 10.0) — VSB Engineering College, Karur, TN **Batch 2024 – 2028**

PROFESSIONAL EXPERIENCE

Full-Stack Development Intern — Neura Global **06/2026 – 07/2026**

- **Campus Recruitment Platform:** Engineered **CampusConnect**, a recruitment management system with 5-stage application funnels and 4-tier Role-Based Access Control (RBAC) security using React 19, TypeScript, Node.js, Express, and Supabase.
- **Real-Time Funnel & Cloud Storage:** Integrated **Socket.IO** for real-time interview status updates and conflict-free scheduling, pairing with an encrypted **Cloudinary** document vault for verified student credential storage.

Software Development Intern — Infosys Springboard **11/2025 – 01/2026**

- **Enterprise Recognition System:** Developed **StarWall** peer recognition and analytics dashboard using Java 17, Spring Boot, modular microservices, and test-driven development (TDD) adhering to agile Git branch workflows.

PROJECTS

VSBEC IT Vault — Academic Task Manager & Compiler Platform | *Top 50 SIH Nominee* **2026**

React 19, TypeScript, Tailwind CSS, Node.js, Express, PostgreSQL, Monaco Editor, Docker [Demo] | [GitHub]

- **Campus Production Adoption:** Deployed and actively used daily by **365+ enrolled students** across 6 sections for assignment submissions, proof verification, and automated LeetCode/GitHub sync.
- **Isolated Compiler Sandbox:** Built multi-language execution runtime (**C, C++, Java 17, Python 3**) with isolated child-process jails, timeout traps, server-side test evaluation, and a 3-tier approval hierarchy.

GOAT Code Editor (GOAT-CE) — Real-Time Collaborative Cloud IDE | *1st Place National Winner* **2026**

React 19.2, TypeScript, Pure WebRTC, Socket.IO 4.8, Monaco Editor Core, PostgreSQL, Piston API [Demo] | [GitHub]

- **Zero-SDK WebRTC Voice Mesh:** Built custom P2P audio mesh handling SDP handshakes, ICE candidates, and hardware echo cancellation without paid 3rd-party SDKs for ultra-low latency collaboration.
- **Sub-Pixel Diff Streaming:** Implemented Socket.IO pipeline streaming operational transformations, remote cursor presence, and live selection across Monaco editor. Won **1st Place (₹5,000)** at Code Thugs 2K26.

Smart Helmet IoT — Rider Safety & Accident Prevention System | *Top 50 SIH Nominee* **2025**

Embedded C/C++, Arduino Nano, RF 433MHz, MQ-3 Alcohol Sensor, IR Proximity, Eye-Blink Sensor, SPDT Relay [GitHub]

- **Dual-Unit Ignition Interlock:** Engineered embedded safety system linking helmet transmitter and bike receiver over RF 433MHz; enforces optical IR wear detection and MQ-3 breath alcohol sensing with **<500ms automated ignition cut-off**.
- **Drowsiness Alert & Fail-Safe Watchdog:** Integrated IR eye-blink eyelid sensor for real-time micro-sleep alerts, implementing a **5s RF fail-safe watchdog** that automatically isolates vehicle ignition upon signal loss or tampering.

AURA — AI Unified Retrieval Assistant (Air-Gapped Local RAG) **2026**

Java 17, Spring Boot 3, React, Vite, Electron, SQLite, ChromaDB, Ollama (LLaMA 3), WebSockets [GitHub]

- **100% Offline RAG Pipeline:** Built enterprise desktop assistant executing local LLMs (Ollama) with loopback network isolation, PDF vector search, and real-time WebSocket token streaming with zero API costs.
- **Cross-Format Intelligence:** Implemented local document parsing, semantic search over enterprise archives, and local speech-to-text processing packaged into an Electron desktop bundle.

HONORS & ACHIEVEMENTS

- **1st Place National Winner** — *Code Thugs 2K26* (National Hackathon Champion for GOAT Collaborative IDE; ₹5,000 prize).
- **National Finalist** — *Odoo × NMIT Bangalore Hackathon 2026* (Qualified for 24-hr Grand Finale at NMIT Bengaluru).
- **Top 50 / 300+ Teams** — *Smart India Hackathon (SIH 2026 & SIH 2025)* (Portal Nominee for VSBEC IT Vault and Smart Helmet IoT).
- **3rd Prize Winner** — *8-Hour AI/ML Hackathon 2026* (Awarded by Dept of AI&ML for FINDORA AI multimodal system).

VERIFIED CERTIFICATIONS

Cloud & AI:	Google Cloud: Prompt Design in Vertex AI, Responsible AI, Intro to LLMs; Tata GenAI (Forage)
Software & Systems:	Infosys Springboard Internship Certificate; Cisco Intro to IoT; TCS iON Career Edge; Salesforce Admin